Pasada

"Know when to ride. Know when to leave."

PROJECT BRIEF

Demand-first public transport coordination platform for jeepney cooperatives · Metro Manila
SparkFest 2026 — GDG PUP · SDG 11: Sustainable Cities & Communities

Field	Detail
Project	Pasada — demand-first public transport coordination platform (B2B SaaS for jeepney cooperatives)
Event	SparkFest 2026, organized by Google Developer Groups on Campus — Polytechnic University of the Philippines (GDG PUP)
SDG alignment	SDG 11 — Sustainable Cities & Communities (primary); SDG 9 — Industry, Innovation & Infrastructure
Submission deadline	July 2, 2026
Repository	github.com/sjmbaldesco/masatohackathon
Status	MVP in development (Day 2 of a 4-day build, as of June 29, 2026)

Executive Summary

The jeepney system runs on a two-sided waiting problem: passengers wait blindly at the curb not knowing when a jeep will arrive, and drivers idle at terminals not knowing where passengers are. Pasada closes that **information gap** — not a technology gap — by digitizing street-level demand and surfacing it to drivers and cooperative dispatchers in real time. Passengers **broadcast** that they are waiting (no booking, no payment, no legal friction); drivers and dispatchers see aggregated demand and make smarter departure and dispatch decisions.

Problem

Both sides of the jeepney system lose time, and cooperatives have no data to dispatch smarter — producing inefficient routes, long waits, and underutilized units. This disproportionately hurts low-income commuters who depend on jeepneys as their primary transport. **This is not a technology gap. It is an information gap.**

Solution

Pasada surfaces the street-level information that already exists — where passengers are waiting, how many, and where they are going. Passengers tap **"I'm waiting here"** to broadcast demand and feed a live heatmap; drivers see aggregated demand along their route and a Departure Confidence Score; cooperatives see the full picture and dispatch optimally. Think Grab — *without* booking a private vehicle: matching demand against an existing public route.

Objectives

Hackathon objectives

- Ship a working MVP across all three surfaces — passenger app, driver dashboard, cooperative dashboard — and submit before the July 2 deadline.
- Demonstrate the **Departure Confidence Score** (the killer feature) live during the pitch.
- Show credible SDG 11 impact paired with an adoption model that actually works.

Product objectives

- Reduce passenger wait time and driver idle time on a pilot route.
- Give cooperatives real-time demand visibility and AI-backed dispatch recommendations.
- Prove the cooperative-enrollment adoption model (the fleet, not the individual driver, is the customer).

Users & Stakeholders

Stakeholder	Role	Primary need
Passenger	Demand broadcaster	Know when a jeep is coming; signal they are waiting
Driver	Supply operator	Know where passengers are before departing
Cooperative dispatcher	Fleet coordinator	Optimize dispatch across all active units
LGU / LTFRB / MMDA	Regulators & partners	Route-efficiency data and compliance insight

Scope

In scope — MVP

- **Passenger app:** select route & stop; view nearest-jeep ETA, occupancy, seats; "I'm waiting here" broadcast; arrival push notification.
- **Driver dashboard:** live demand heatmap, seat-queue status, Departure Confidence Score; GPS auto-updates every 5–10s.
- **Cooperative dashboard:** route overview, live KPIs, Gemini-powered dispatch recommendations, historical analytics.
- **Live demand heatmap** aggregated from broadcasts (no individual pins); Google sign-in.

Out of scope — what Pasada is NOT

- Not ride-hailing (no guaranteed pickup, no payment) · Not seat booking/reservation.
- Not a replacement for cooperative dispatch — it augments it.
- Not dependent on every driver installing the app — cooperative enrollment solves adoption.

Future scope: SMS / stop-kiosk fallback for inclusion; fare/boundary digitization; expansion to other metros and to UV Express & buses.

Key Features

Feature	What it does
Passenger app	Route + stop selection, real-time ETA / occupancy / seats, demand broadcast, ~2-minute arrival alert
Driver dashboard	Demand heatmap per stop, seat-queue status, Departure Confidence Score

Cooperative dashboard	Route queue & active units, KPIs (wait time, occupancy, revenue/unit), Gemini dispatch insights
Live demand heatmap	Color-coded intensity per stop from broadcasts — 🔴 high 🟠 medium 🟢 low — updated via Firestore listeners

Killer feature — Departure Confidence Score

Instead of telling a driver to "wait until full," Pasada estimates whether departing *now* is optimal — from passengers already waiting, historical boarding patterns, live traffic (Google Maps), and current app demand. It reframes the product from "find my jeepney" to "optimize the whole route," addressing every stakeholder's incentive at once.

```
Departure Confidence        92%
Expected passengers:        17-18
Expected travel time:       41 min
Expected revenue:           PHP 320
[ Recommended: Depart Now ]
```

Technical Approach

Layer	Technology
Frontend	React 18 + Vite + Tailwind CSS
Backend	FastAPI (Python) on Google Cloud Run
Database / real-time	Firebase Firestore with real-time listeners
Auth	Firebase Authentication (Google login)
Maps & routing	Google Maps Platform — Maps JS, Directions, Distance Matrix, Geocoding
AI	Gemini API (decision support, not chat)
Notifications	Firebase Cloud Messaging
Hosting	Cloud Run (backend) + Firebase Hosting (frontend)

Flow: passenger/driver apps → Google Maps API → FastAPI backend → Firebase Auth + Firestore + Gemini → cooperative dashboard. Core Firestore collections: *drivers, passengers, routes, stops, queue, requests, cooperatives.*

Success Metrics

Hackathon (judging-aligned)

- Working demo of all three surfaces with the Departure Confidence Score running live.
- Deployed end-to-end (Cloud Run + Firebase Hosting) with a clear, credible SDG 11 impact narrative.

Product KPIs

- Average passenger wait time & driver idle time per unit (down).
- Average occupancy / load factor and revenue per unit (up).
- Demand broadcasts per day and share of dispatch recommendations acted on.

Timeline & Milestones

Day	Milestones
Day 1 — Jun 28	Finalize concept & roles; GitHub repo & branches; init FastAPI backend; set up Firebase (Firestore + Auth); wireframe all three UIs
Day 2 — Jun 29 ● current	Backend auth, GPS-update & demand-broadcast endpoints; passenger app + driver dashboard scaffolds with Google Maps; Firestore real-time listeners wired; begin cooperative dashboard
Day 3 — Jun 30	Departure Confidence Score logic; Gemini dispatch integration; heatmap on passenger & driver maps; Cloud Messaging arrival alerts; full integration test
Day 4 — Jul 1	QA & bug fixes; deploy backend (Cloud Run) & frontend (Firebase Hosting); finalize README & pitch deck; rehearse demo; submit before Jul 2

Risks & Mitigations

Risk	Mitigation
"Why would every driver install this?" (adoption)	Cooperatives enroll the whole fleet; drivers get the app as part of membership — adoption is top-down, not one-by-one
Cold start / sparse passenger data	Driver-side value (seat queue, traffic-aware ETA) works even at low passenger density; seed a single pilot route first
Google Maps API cost at scale	Cache & batch requests, compute ETA server-side, call Distance Matrix sparingly
Connectivity & low-end devices / inclusion	Lightweight app; planned SMS / stop-kiosk fallback for non-smartphone commuters
Gamed or stale demand signals	Aggregate (no individual pins); auto-expire broadcasts on jeep arrival
Regulatory friction	Position as a compliance partner — share route-efficiency data with LTFRB / LGUs
4-day build risk	Scope to MVP; keep seeded/mock demand data as a demo fallback

SDG Alignment

SDG / target	How Pasada addresses it
11.2 — safe, affordable, accessible transport	Reduces wait time and improves PUV efficiency for urban commuters
11.b — inclusive, sustainable urbanization	Gives cooperatives and LGUs data for evidence-based transport decisions
9.1 — resilient infrastructure	A digital infrastructure layer over an existing analog transport system

Team & resources: Built for SparkFest 2026 — GDG PUP. Repo: github.com/sjmbaldesco/masatohackathon · Google Maps Platform, Firebase, Gemini API, FastAPI.